

“THE BEST THAT MOST OF US CAN HOPE TO ACHIEVE IN **PHYSICS** IS SIMPLY TO **MISUNDERSTAND AT A DEEPER LEVEL.**”

Wolfgang Pauli, Austrian physicist, in a letter to Jagdish Mehra in Berkeley, California, 1958

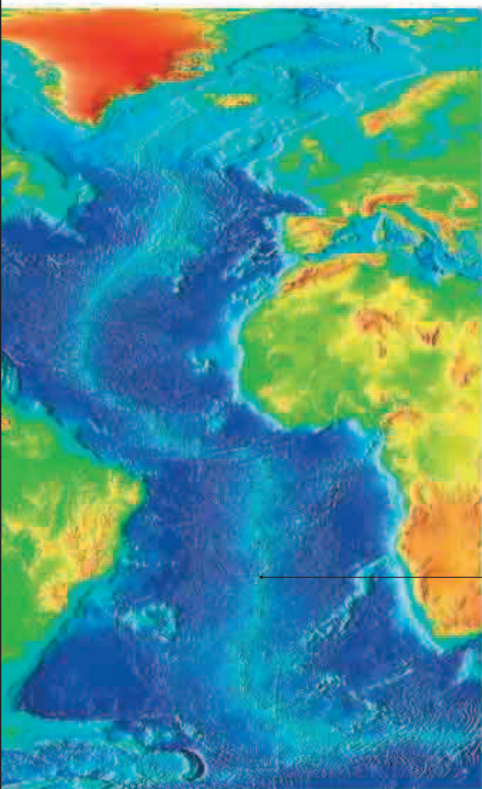
SCOTTISH ENGINEER JOHN

LOGIE BAIRD (1888–1946) had been experimenting with transmitting moving images and now, using a semimechanical device, he achieved it. In March, **television was demonstrated** in a London department store. At first images were silhouettes, but by October, he was **transmitting pictures with grayscale** made up of 30 vertical lines running at five pictures per second.

In May, **John Scopes** (1900–70), of Tennessee, was arrested for teaching Darwin's theory of evolution, which was prohibited in the state's schools. He was tried, found guilty, and fined \$100.

Quantum physics had shown how the electrons of an atom existed in fixed energy levels, but Austrian physicist **Wolfgang Pauli** (1900–58) went further with his **Exclusion Principle** saying that no two particles could occupy the same quantum state at the same time. Later, two categories of particles would be distinguished—matter-associated particles (fermions) that obeyed Pauli's principle, and force-associated particles (bosons) that did not (see 1974).

The German Atlantic Meteor Expedition began to survey the ocean's floor and **discovered an unbroken ridge running from north to south.**



Atlantic Ridge

Mid-Atlantic ridge

The complementary shapes of Atlantic coastlines suggest that they were once joined together. This ridge shows where new seafloor has pushed them part.

March 25 John Logie Baird publicly demonstrates television

June 1 Danish hygienists **Louis Fredericia and Eiler Holm** show that vitamin A deficiency causes night blindness

Late 1925 German Atlantic Meteor Expedition discovers mid-Atlantic ridge

May 5 John Scopes is arrested; his trial begins on July 10

July Canadian Chemist **Walter Bloor** describes a classification of fats and oils, and coins the term **lipid**

November Wolfgang Pauli explains his **Exclusion Principle**



Two ventriloquist's dolls became the first TV personalities when John Logie Baird demonstrated his television system to scientists of the Royal Institution and members of the press.

IN JANUARY, JOHN LOGIE BAIRD

was back showing off his improved television. This time it was to members of the Royal Institution in London, UK. He now achieved **clearer images with tonal gradation of grays.**

Although the images were still blurred, Baird had increased the picture frequency, so **the movements were smoother.**

Austrian physicist **Erwin Schrodinger** (1887–1961) examined earlier suggestions that matter was both particle-like and wavelike in nature. Schrodinger calculated a particle's distribution of energy in space—its wave function—and urged that the wave idea alone was critical to understanding reality. His work would be the **basis for quantum mechanics.**

Six years after the *New York Times* had ridiculed his suggestion that **a rocket was a possibility**, American physicist Robert Goddard (1882–1945) launched one. Using his specially



developed liquid fuel, **Goddard's rocket reached a height of 40 ft (12m)** in less than three seconds.

In August, American chemist **James Sumner** (1887–1955) made a breakthrough in biochemistry. By grinding up jack beans, he had **isolated crystals**

The first rocket

Robert Goddard launched the first liquid-fuel rocket on his aunt's farm in Massachusetts. Three people witnessed the event: his wife, his machinist, and a fellow physicist.

of the enzyme urease.

Enzymes are substances found in living tissues that speed up the chemical reactions of metabolism. When Sumner analyzed his crystals, he found that they were protein. In this way, he had **demonstrated that biological enzymes were proteins.**

In December, another American chemist, Gilbert Lewis (1875–1946), **coined the term photon** for a unit of radiant energy. It came to be used to describe a particle of light energy.

PVC, polyvinyl chloride (often called vinyl), **made its first**

QUANTUM THEORY

The introduction of quantum theory was a pivotal point in early 20th century physics that began with Max Planck's explanation of black body radiation (see 1900). The theory is based on the idea that energy exists in discrete parcels, or quanta (from the Latin *quantus*, meaning “how much”). These parcels equate to the fixed quantities of energy

absorbed or emitted by atoms. This is realized as specific wavelengths of radiation, each associated with particular quanta of energy: red light wavelength is transmitted by low-energy parcels and blue light by high-energy ones. Danish physicist Niels Bohr explained this in terms of the atom's electrons moving between energy levels, or orbits.

January 26 John Logie Baird publicly demonstrates improved version of his television

February 7 Enrico Fermi explains how fundamental particles of matter obey the **Exclusion Principle**

January Erwin Schrodinger develops his mathematical formulation of **quantum-wave mechanics**

March 16 Robert Goddard launches the first liquid-fuel rocket

August 26 British physicist **Paul Drac** independently gives Fermi's explanation of fundamental particles

August James Sumner describes isolating the enzyme urease and shows that **enzymes are proteins**